

## Confidence, Joint Distributions and Errors

### Exact conditional calculations; no empirical confidence in method rankings.

This v2 replaces v1’s independently assigned method means and error assumptions. It gives a mathematically coherent *conditional reference*, not a fitted prediction of RARE or any baseline. Scientists can test metric relationships and error patterns against it, while recording which assumptions fail.

### Confidence audit: what the existing data identify

| Quantity               | Existing support                            | Confidence scope                |
|------------------------|---------------------------------------------|---------------------------------|
| API provenance         | 8 complete calls, one PIPE3 root            | High for recorded calls         |
| Core update time       | 0.520 / 0.297 ms; one per arm               | Exact events; no p95 forecast   |
| Serialized state       | 3,408 / 4,688 bytes                         | Exact snapshots; no future size |
| Future method effect   | No post-update task executed                | Not identified                  |
| Brier / calibration    | No qualified probability/outcome population | Not identified                  |
| Drift / forgetting     | No matched live drift streams               | Not identified                  |
| Method/root covariance | No independent paired future histories      | Not identified                  |

All three live arms chose the same peer and reused the same artifact. Their producer, recipient and old adoption checks were 2/3, 3/3 and 1/2. These are check fractions, not success-rate estimates. The source updates do not implement the active method’s validated future-credit contract. Numerical precision in a guide cannot raise confidence in unobserved performance.

### One joint model, one selected outcome

In each task, candidate  $H$  has assumed success probability .7 and  $L$  has .3. Task context swaps their peer identities. These are latent *evaluation* roles, never oracle information supplied to the selector. Reduced-form routing skill  $\kappa$  means probability of choosing  $H$  is  $(1 + \kappa)/2$ . For the chosen peer only,  $Y \in \{0, 1\}$  is an independent terminal outcome.

$$Q = \mathbb{E}[Y] = .5 + .2\kappa, \quad C = \frac{1 + h + (1 - Y)}{1.5} + \xi, \quad U = Y - .1C.$$

Case:  $h = .02$  is assumed policy overhead; failure adds one repair-cost unit. The baseline expected cost is 1.5 units. Jitter  $\xi$  takes  $\pm .1$  normalized cost units with equal probability, independently of  $Y$ . Thus quality and cost are related through failure, rather than separate invented columns. All these parameters are *unestimated assumptions*.

**Table 1 counterpart: coherent conditional scenarios, not named methods**

| Scenario | $Q$   | Brier  | Population ECE | $\mathbb{E}[C]$ | $\mathbb{E}[U]$ |
|----------|-------|--------|----------------|-----------------|-----------------|
| S0       | 0.500 | 0.2500 | 0.000          | 1.000           | 0.400           |
| S1       | 0.500 | 0.2100 | 0.000          | 1.013           | 0.399           |
| S2       | 0.550 | 0.2125 | 0.050          | 0.980           | 0.452           |
| S3       | 0.600 | 0.2125 | 0.050          | 0.947           | 0.505           |
| S4       | 0.600 | 0.2325 | 0.150          | 0.967           | 0.503           |

S0: random routing and constant .5 forecast. S1: routing unchanged, hypothetical perfectly calibrated forecast. S2:  $\kappa = .25$ , forecast bias .05. S3:  $\kappa = .50$ , same bias. S4:  $\kappa = .50$ , bias .15, overhead .05 instead of .02. Forecasts  $p = q_{\text{chosen}} + e$  are hypothetical calibration scenarios using the model’s latent  $q$ , not an implementable oracle predictor or evidence of judgment information. Forecast skill alone improves Brier in S1 but not task quality. S3 improves routing over S2 without automatically improving Brier or ECE. No real method is assigned an S-label or  $\kappa$ .

**Distribution and error matrix: S3, 30 iid frozen-policy episodes****Exact conditional calculations; no empirical confidence in method rankings.**

| Stream statistic | Mean   | SD     | Central 95% prediction interval | Mass (%) |
|------------------|--------|--------|---------------------------------|----------|
| Quality          | 0.6000 | 0.0894 | [0.4333, 0.7667]                | 96.2     |
| Brier            | 0.2125 | 0.0383 | [0.1405, 0.2905]                | 95.1     |
| Sample ECE       | 0.0999 | 0.0559 | [0.0167, 0.2300]                | 95.9     |
| Cost             | 0.9467 | 0.0624 | [0.8267, 1.0711]                | 95.3     |
| Utility          | 0.5053 | 0.0954 | [0.3256, 0.6838]                | 95.1     |

These are *exact finite-count distributions for the iid frozen-policy model and specified Rademacher jitter*, not confidence intervals for empirical method parameters. Quality comes from success counts. Brier and sample ECE use the *same* selected labels and forecasts; cost uses the same failures plus specified jitter; utility uses those same quality and cost records. No independent error bars are assigned. Mass is the model's probability inside the unrounded endpoints; discreteness can make it exceed 95%. It is not measured holdout coverage.

**Dependence matrix for the same conditional stream**

| Correlation | $Q$     | Brier   | ECE     | Cost    | Utility |
|-------------|---------|---------|---------|---------|---------|
| $Q$         | 1.0000  | -0.6124 | -0.5386 | -0.9562 | 0.9998  |
| Brier       | -0.6124 | 1.0000  | 0.5093  | 0.5855  | -0.6123 |
| ECE         | -0.5386 | 0.5093  | 1.0000  | 0.5150  | -0.5385 |
| Cost        | -0.9562 | 0.5855  | 0.5150  | 1.0000  | -0.9616 |
| Utility     | 0.9998  | -0.6123 | -0.5385 | -0.9616 | 1.0000  |

Utility is derived, not an extra independent endpoint. The covariance matrix is positive semidefinite but includes this linear dependence. Do not invert it as though all five endpoints were independent. Cost/quality correlation here follows an assumed failure-repair model; the real correlation must be measured and can differ.

$$\text{Var}(\bar{U}) = \text{Var}(\bar{Q}) + .1^2 \text{Var}(\bar{C}) - .2 \text{Cov}(\bar{Q}, \bar{C}).$$

Case: the negative quality–cost covariance widens utility variability relative to an independence calculation, because a failure reduces quality and adds repair cost simultaneously.

**How the prediction and error distributions are calculated**

For a chosen type with success rate  $q$  and fixed forecast  $p$ ,

$$\mathbb{E}[(p - Y)^2] = q(1 - q) + (p - q)^2.$$

Case: S3 selects .7 with probability .75 and .3 with probability .25; forecasts are .75 and .35. Both conditional Brier means are  $.21 + .05^2 = .2125$ . The resulting selected success rate is .60, not the .50 prevalence of an unselected balanced population.

For  $N = 30$ , integrate  $N_H \sim \text{Bin}(30, .75)$ , followed by  $S_H \sim \text{Bin}(N_H, .7)$  and  $S_L \sim \text{Bin}(30 - N_H, .3)$ . Then

$$\widehat{\text{ECE}} = \frac{|.75N_H - S_H| + |.35(30 - N_H) - S_L|}{30}.$$

Case: each occupied forecast bin contributes its absolute forecast/count error; empty bins contribute zero. Finite-sample ECE has an upward absolute-value bias, so its expected sample value exceeds population ECE .05. For S0 both forecast types merge into one .5 bin; treating them as two bins would be incorrect. The calculation enumerates binomial counts and jitter counts exactly. It does not generate fake agent predictions, real API labels, or synthetic paper results.

**Paired errors: outcome coupling cannot be assumed away****Exact conditional calculations; no empirical confidence in method rankings.**

Compare S3 with S2 on the same frozen task/quality population. Couple route draws for this analytical sensitivity only. The chosen peer differs with probability  $D = (.50 - .25)/2 = .125$ , and  $\Delta Q = 0.050$ ,  $\Delta U = 0.0533$  when overhead is equal. Let  $u = \Pr(Y_H = 1, Y_L = 1)$ ; its feasible range is  $[0, .3]$ .

$$\text{Var}(\Delta Y) = .125(1 - 2u) - .05^2, \quad \text{Var}(\Delta U) = (1 + .1/1.5)^2 \text{Var}(\Delta Y) + 2(.1)^2(.1)^2.$$

Case: the final term is specific to independent Rademacher cost jitters. Correlated jitters require the covariance term; merely bounded jitter does not justify this equality. The potential outcomes are evaluation assumptions; unselected outcomes never enter learning feedback.

| Assumed $u$ | Paired episode quality variance | 30-episode paired utility SD |
|-------------|---------------------------------|------------------------------|
| 0.00        | 0.1225                          | 0.0682                       |
| 0.21        | 0.0700                          | 0.0516                       |
| 0.30        | 0.0475                          | 0.0425                       |

Same marginal quality improvement can have materially different paired errors. The shared-potential-outcome assumption applies only to this frozen-policy one-step model. Real adaptive arms have independent histories and may change candidate artifacts, tasks, costs and missingness; the same RNG seed does not make their realized outcomes paired counterfactuals.

**Three body tables: what can receive empirical confidence**

| Paper matrix    | Joint quantities required                                      | Current forecast status                     |
|-----------------|----------------------------------------------------------------|---------------------------------------------|
| 1: methods      | $Y, p, a, C$ , root and policy-history identity                | All eight method rows unidentifiable        |
| 2: online/drift | Full service times, arrivals, retained state, old/new holdouts | Core anchors only; tails/adaptation unknown |
| 3: mechanisms   | Four-cell joint stream outcomes; independent owner truth       | Interaction and ungated rate unknown        |

An unqualified published adapter must not be replaced by guessed values for a strong control. Choice propensity is not predicted success probability. Store the two separately. Legal varied judgments and count-only/J-masked controls are necessary before attributing information gains to judgment content.

**Interaction and safety targets must describe observable relationships**

For evidence/credit cells, define  $I_U = U_{11} - U_{10} - U_{01} + U_{00}$ . Its variance is  $c^\top \Sigma c$  with  $c = (1, -1, -1, 1)$  and the cells' joint stream covariance; four independent guessed SDs cannot replace  $\Sigma$ . In this model  $I_U = (1 + .1/1.5)(.2)I_\kappa - (.1/1.5)I_h$ . Case: additive routing and overhead imply zero interaction. A utility interaction target .02 with  $I_h = 0$  requires  $I_\kappa = 0.09375$ ; it is a testable skill requirement, not an already predicted gain.

Fail-closed producer credit is a protocol requirement. Its real violation rate and owner-oracle coverage need independent negative cases, UNKNOWN denominators and root/stream-aware error analysis. V1's .25/.15/.10/.05 ungated rates are withdrawn as unsupported expectations. A core timing of .297 ms also cannot identify p95. A 5 ms maximum envelope would drain four events in 20 ms, but that is a conditional queue implication; full service and tail measurements remain required, including backbone/retrieval and publish/read costs.

## Root and history variation can overturn apparent precision

### Exact conditional calculations; no empirical confidence in method rankings.

Use  $u = .21$  only as an independent-potential-outcome sensitivity point. Let  $\rho$  be the ICC of *paired episode utility differences*; for 30 exchangeable episodes,  $DE = 1 + 29\rho$  and  $\sigma_{\text{stream}}^2 = \text{Var}(\Delta U)DE/30$ . With three roots and six independent streams per root,

$$\text{SE}^2(\overline{\Delta U}) = \frac{\sigma_{\text{root}}^2}{3} + \frac{\sigma_{\text{stream}}^2}{18}.$$

Case: stream variance already incorporates episodes and dependence; do not divide by 30 again. Root variance reflects differing method effects across tasks; increasing episodes within three roots cannot remove it.

| Assumed $\rho$ | Assumed root SD | Stream SD | Conditional normal 95% planning band |
|----------------|-----------------|-----------|--------------------------------------|
| 0.00           | 0.00            | 0.0516    | [0.030, 0.077]                       |
| 0.00           | 0.02            | 0.0516    | [0.020, 0.086]                       |
| 0.00           | 0.05            | 0.0516    | [-0.008, 0.115]                      |
| 0.05           | 0.00            | 0.0808    | [0.016, 0.091]                       |
| 0.05           | 0.02            | 0.0808    | [0.010, 0.097]                       |
| 0.05           | 0.05            | 0.0808    | [-0.014, 0.121]                      |
| 0.10           | 0.00            | 0.1019    | [0.006, 0.100]                       |
| 0.10           | 0.02            | 0.1019    | [0.001, 0.106]                       |
| 0.10           | 0.05            | 0.1019    | [-0.020, 0.127]                      |

Every row has the *same* assumed mean effect, yet bands differ and can cross zero. These are known-variance planning approximations, not real intervals, power guarantees or fitted confidence. Three roots cannot reliably estimate a root-effect distribution; actual root-aware inference and small-sample critical values must be specified before confirmation. No measured variance is available.

### What scientists should collect before calibrating a forecast

1. Qualify structural roots, scorer/owner truth and baselines first. Freeze predicted-success labels separately from propensities and channel eligibility.
2. In a bounded real development pilot, preserve root/stream/task identity, before-selection state, predictions, choice probabilities, diverse legal J, actual action, independent  $Q_p/Y$ , complete costs, timestamps and UNKNOWN. Each arm has its own realized history.
3. Estimate means *and* joint stream covariance, quality–cost dependence, paired differences, autocorrelation, missingness and cross-root heterogeneity. A fixed all-accept source cannot identify judgment-content information.
4. Fit prospective distributions using development data only. Validate on future held-out streams/roots: bias, 80%/95% interval coverage and width, calibration and negative/failed cases. Insufficient coverage means the forecast lacks confidence even if its mean happens to match.
5. Freeze confirmation estimands, roots, budgets and stopping rules after the forecast is calibrated. Do not manufacture narrow errors or lower the Goal to make a negative result agree with a guide.

Until those observations exist, method means, rankings and error distributions remain unestimated. The equations and count laws have checked conditional validity; they do not certify that the assumed world describes the benchmark. This guide changes neither the method nor the acceptance standards, and uses zero new API calls or GPU jobs. All v1 artifacts remain archived.